

Introduction to IoT

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Introduction

- **Internet of Things (IoT)** comprises things that have unique identities and are connected to the internet.
- **Existing devices** , such as networked computers or 4G enabled mobile phones already have some form of unique identities and are also connected to the internet, the focus on IoT is in the configuration, control and networking via the internet of devices or things , that are traditionally not associated with the Internet.
- These include devices such as thermostats, utility meters, a blue tooth- connected headset, irrigation pumps and sensor or control circuits for an electric car's engine
- Experts already forecasted that by the year 2020 there will be a total of **50 billion devices/ things** connected to the internet.
- The scope of IoT is not limited to just connected things(Devices, appliance, machines) to the Internet.
- Applications on IoT networks extract and create information from lower level data **by filtering, processing , categorizing, condensing and contextualizing the data.**
- The information obtained is then organized and structured to infer knowledge about the system and or its user, its environment and its operations and progress towards its objectives, allowing a smarter performance.

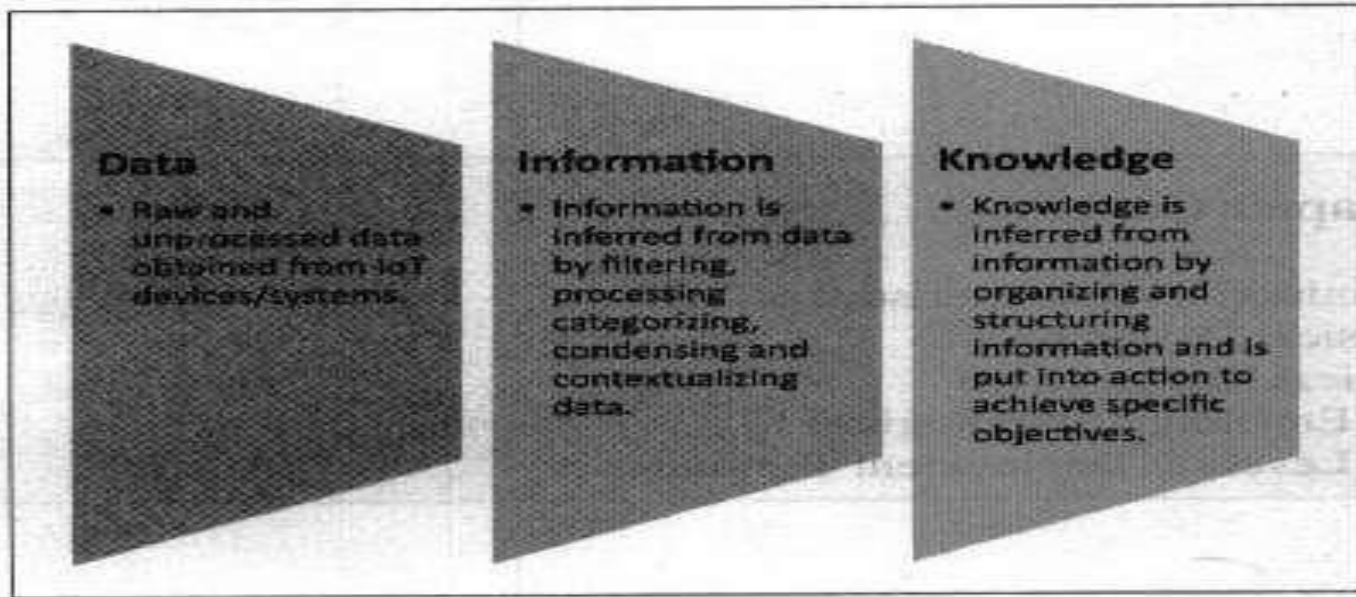


Figure 1.1: Inferring information and knowledge from data

The Internet of Things (IoT) describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

“**The Internet of Things (IoT)** is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.”

- Internet of Things (IoT) has been defined as

Definition: A dynamic global network infrastructure with selfconfiguring capabilities based on standard and interoperable communication protocols where physical and virtual Things have identities, physical attributes, and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network, often communicate data associated with users and their environments.

IoT Characteristics

Dynamic & Self-Adapting: IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context, or seed environment. For example, consider a surveillance system comprising of a number of surveillance cameras. The surveillance cameras can adapt their modes (to normal or infra-red night modes) based on whether day or night. Cameras could switch from lower resolution to higher resolution modes when any motion is detected and alert nearby cameras to do the same.

In this example, the surveillance system is adapting itself based on the context and changing (e.g.dynamic) conditions.

Self-Configuring: IoT devices may have self-configuring capability, allowing a large number of devices to work together to provide certain functionality (such as weather monitoring). These devices have the ability configure themselves (in association with the IoT infrastructure), setup the networking, and fetch latest software upgrades with minimal manual or user intervention.

Interoperable Communication Protocols: IoT devices may support a number of Interoperable communication protocols and can communicate with other devices and with the infrastructure.

Unique Identity: Each IoT device has a unique identity and a unique identifier (such as an IP address or a URI). IoT systems may have intelligent interfaces, which adapt based on the context, allow communicating with users and the environmental contexts.

IoT device interfaces allow users to query the devices, monitor their status, and control them remotely, in association with the control, configuration and management infrastructure.

Integrated into Information Network: IoT devices are usually integrated into the information network that allows them to communicate and exchange data with other devices and systems.

IoT devices can be dynamically discovered in the network, by other devices and/or the network, and have the capability to describe themselves (and their characteristics) to other devices or user applications.

For example, a weather monitoring node can describe its monitoring capabilities to another connected node so that they can communicate and exchange data.

Integration into the information network helps in making IoT systems "smarter" due to the collective intelligence of the individual devices in collaboration with the infrastructure. Thus, the data from a large number of connected weather monitoring IoT nodes can be aggregated and analyzed^{9/9/2023} to predict the weather.

Advantages of IoT

- Internet of things facilitates several advantages in our daily lives. Some of its advantages are given below:
- **Minimize human effort:** As IoT devices interact and communicate with each other, they can automate the tasks helping to improve the quality of a business's services and reducing the need for human intervention.
- **Save time:** By reducing the human effort, it saves a lot of our time. Saving time is one of the primary advantages of using the IoT platform.
- **Enhanced data collection:** Information is easily accessible, even if we are far away from our actual location, and it is updated frequently in real-time. Hence these devices can access information from anywhere at any time on any device.

- **Improved security:** If we have an interconnected system, it can assist in the smarter control of homes and cities through mobile phones. It enhances security and offers personal protection.
- **Efficient resource utilization:** We can increase resource utilization and monitor natural resources by knowing the functionality and how each device works.
- **Reduced use of other electronic equipment:** Electric devices are directly connected and can communicate with a controller computer, such as a mobile phone, resulting in efficient electricity use. Hence, there will be no unnecessary use of electrical equipment.
- **Use in traffic systems:** Asset tracking, delivery, surveillance, traffic or transportation tracking, inventory control, individual order tracking, and customer management can be more cost-effective with the right tracking system using IoT technology.
- **Useful for safety concerns:** It is helpful for safety because it senses any potential danger and warns users. For example, GM OnStar is an integrated device that identifies a car crash or accident on the road. It immediately makes a call if an accident or crash is found.
- **Useful in the healthcare industry:** Patient care can be performed more effectively in real-time without needing a doctor's visit. It gives them the ability to make choices as well as provide evidence-based care.

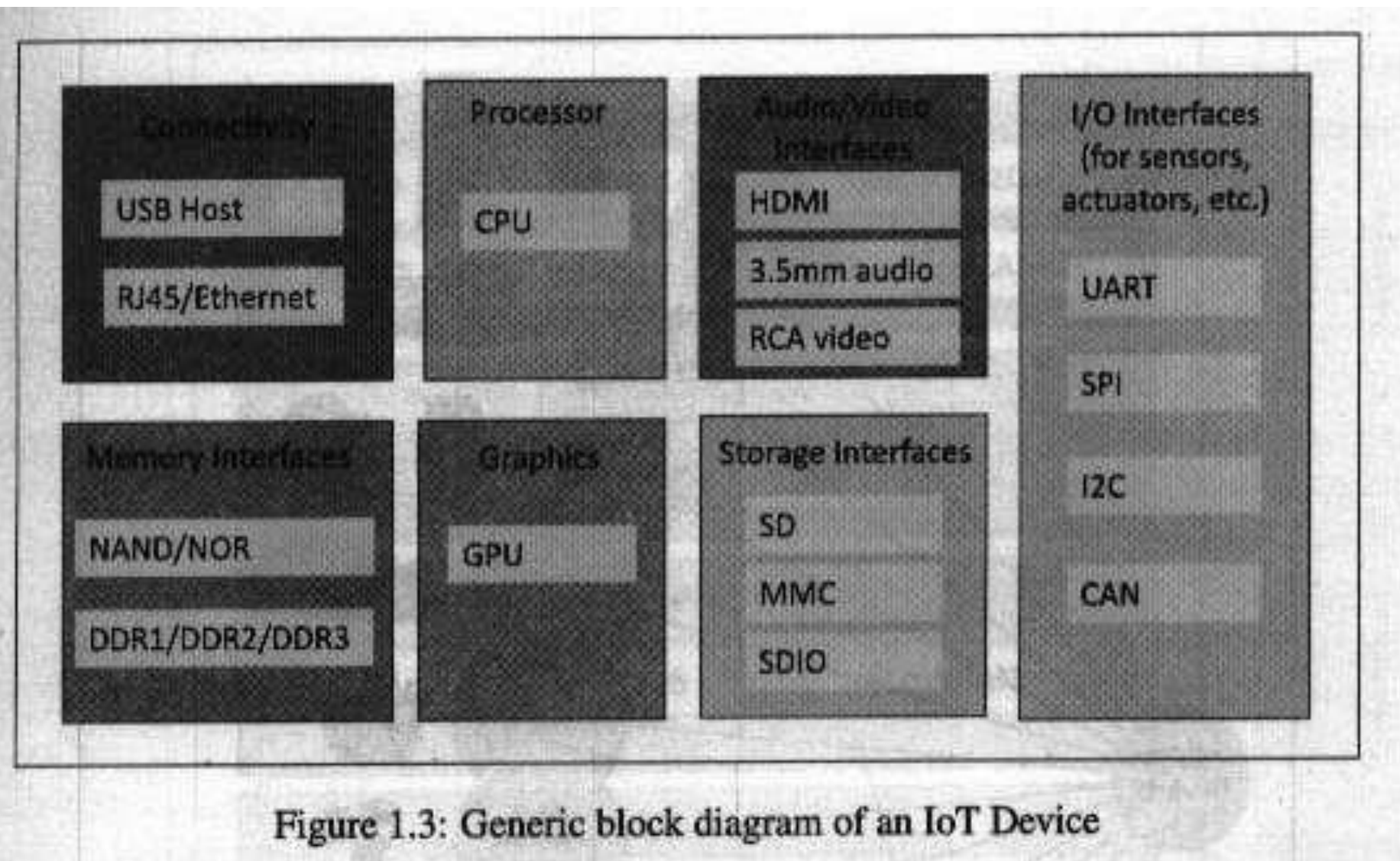
Advantages	Disadvantages
Minimizes the human work and effort	Increased privacy concerns
Saves time and effort	Increased unemployment rates
Good for personal safety and security	Highly dependent on the internet
Useful in traffic and other tracking or monitoring systems	Lack of mental and physical activity by humans leading to health issues.
Beneficial for the healthcare industry	Complex system for maintenance
Improved security in homes and offices	Lack of security
Reduced use of many electronic devices as one device does the job of a lot of other devices	Absence of international standards for better communication

Disadvantages of IoT

- As the Internet of things facilitates advantages, it also creates a significant set of drawbacks. Some of the IoT disadvantages are given below:
- **Security issues:** IoT systems are interconnected and communicate over networks. So, the system offers little control despite any security measures, and it can lead to various kinds of network attacks.
- **Privacy concern:** The IoT system provides critical personal data in full detail without the user's active participation.
- **Increased unemployment:** Unskilled workers or even the skilled ones are at a high risk of losing their jobs, leading to high unemployment rates. Smart surveillance cameras, robots, smart ironing systems, smart washing machines, and other facilities are replacing the humans who would earlier do these works.
- **The complexity of the system:** The designing, developing, maintaining, and enabling the extensive technology to IoT system is quite complicated.
- **High chances of the entire system getting corrupted:** If there is a bug in the system, it is possible that every connected device will become corrupted.
- **Lack of international standardizations:** As there is no international standard of compatibility for IoT, it is problematic for devices from different manufacturers to communicate with each other.
- **High dependency on the internet:** They rely heavily on the internet and cannot function effectively without it.
- **Reduced mental and physical activity:** Overuse of the internet and technology makes people ignorant because they rely on smart devices instead of doing physical work, causing them to become lethargic and inactive.

Physical Design of IoT

- The “Things” in IoT usually refers to IoT devices which have unique identities and can perform remote sensing, Actuating and monitoring capabilities.
- IoT devices can exchange data with other connected devices and applications (directly or indirectly), or collect data from other devices and process the data locally or send the data to Centralized servers or cloud based applications back ends for processing the data or from some task locally and other task within the IoT infrastructure, based on temporal and space constraints (ie : Memory, processing calibrators, communication latencies and speed and deadlines).
- An IoT device may consist of several interfaces connections to other devices, both wired and wireless. See fig 1.3.
- These include
 - I) IoT interfaces for sensors
 - II) Interfaces for internet connectivity
 - III) Memory and storage interfaces
 - IV) Audio video interfaces.
- An IoT Device can collect various types of data from the on board or attached sensors, such as temperature, humidity, and light intensity.



- **IoT devices** can also be varied types, for instance, wearable sensors, smart watches, LED light automobiles and industrial machines.
- Almost all IoT devices generate data in Some form or the other which when processed by Data Analytics systems leads to Useful information to guide further actions locally or remotely.
- **For instance** sensor data generated by a soil moisture monitoring device in a Garden when processed can help in determining the optimum watering schedule. Fig.1.4 shows different IoT devices.

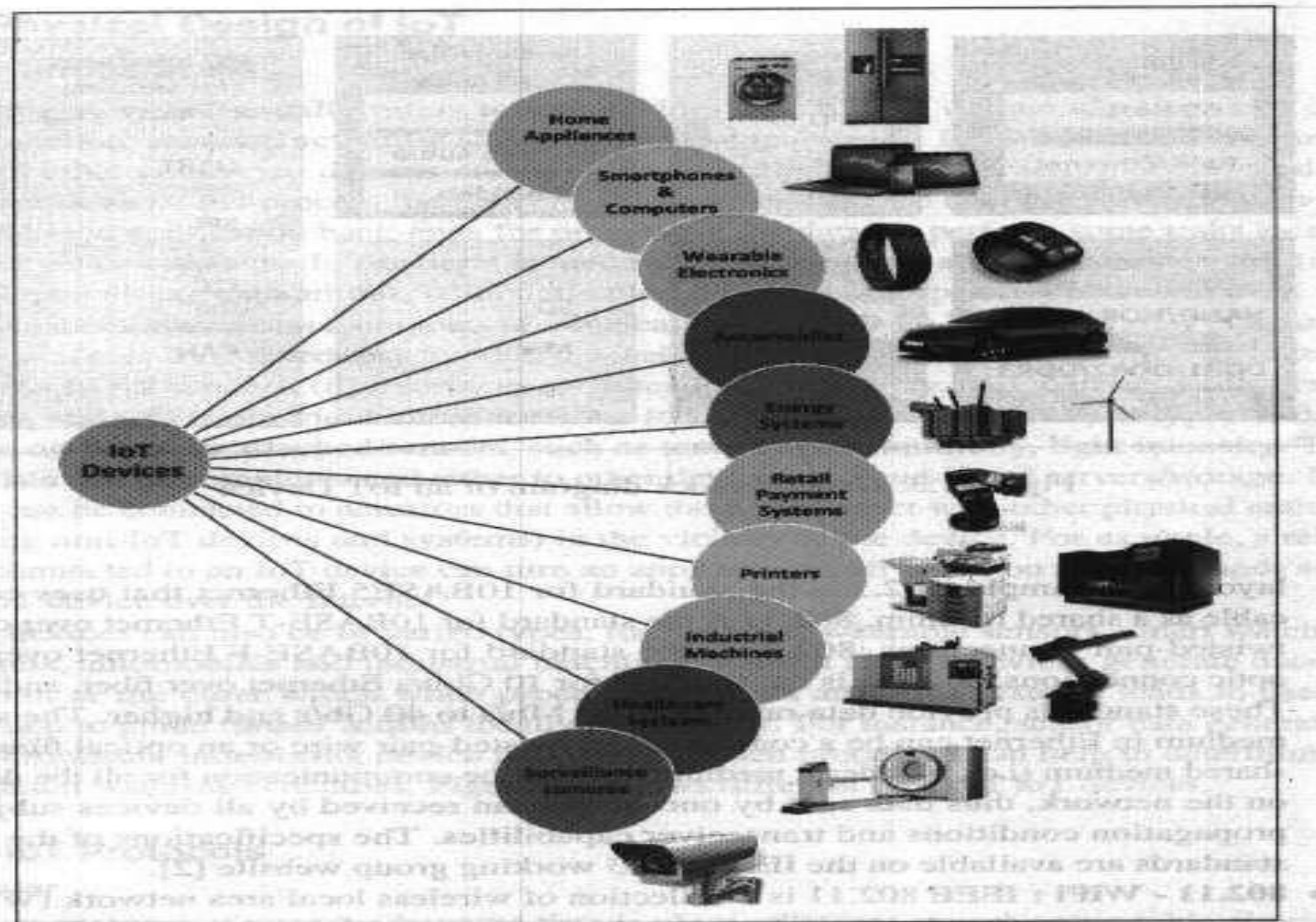


Figure 1.4: IoT Devices

IoT Devices

- 1. Consumer electronics:** Camcorders, cameras, etc.
- 2. Household appliances:** Television, DVD players, washing machine, fridge, microwave oven, etc.
- 3. Home automation and security systems:** Air conditioners, sprinklers, intruder detection alarms, closed circuit television cameras, fire alarms, etc.
- 4. Automotive industry:** Anti-lock breaking systems (ABS), engine control, ignition systems, automatic navigation systems, etc.
- 5. Telecom:** Cellular telephones, telephone Switches, handset multimedia applications, etc.
- 6. Computer peripherals:** Printers, scanners, fax machines, etc.
- 7. Computer networking systems:** Network routers, switches, hubs, firewall etc.
- 8. Healthcare:** Different kinds of scanners, EEG, ECG machines etc.
- 9. Measurement & Instrumentation:** Digital multi meters, digital Oscilloscopes, logic analyzers, PLC systems, etc.
- 10. Banking & Retail:** Automatic teller machines (ATM) and currency counters, point of sales (POS)
- 11. Card Readers:** Barcode, smart card readers, hand held devices etc.

IoT Protocol

Link Layer:

- Link Layer protocols determine how the data is physically sent over the network's physical layer or medium (example copper wire, electrical cable, or radio wave).
- The Scope of The Link Layer is the Local Network connections to which host is attached.
- Host on the same link exchange data packets over the link layer using the link layer protocol.
- Link layer determines how the packets are coded and signaled by the hardware device over the medium to which the host is attached (such as Coaxial Cable).

802.3-Ethernet:

- IEEE 802.3 is a collections of wired Ethernet standards for the link layer.
- For example 802.3 is a standard for 10BASE5 Ethernet that uses coaxial cable as a shared medium,
- 802.3.i is standard for 10BASE-T Ethernet over copper twisted pair connection, 802.3.j is standard for 10BASE-F Ethernet over fiber optic connection
- 802.ae is the standard for 10Gbits/s Ethernet over fiber, and so on.

- These standards provides data rate from 10 Mb/s to 40 gigabits per second and the higher.
- The shared medium is Ethernet can be a coaxial cable, twisted pair wire or and Optical fiber. The Shared medium (i.e. broadcast medium) carries the communication for all the devices on the network.

802.11- Wi-Fi:

- IEEE 802.11 is a collection of wireless Local area network (WLAN) communication standards, including extensive descriptions of the link layer.
- For example 802.11a operate in the 5 GHz band, • 802.11b and 802.11g operate in the 2.4 GHz band.
- 802.11n operates in the 2.4/5GHz band.
- 802.11ac operates in the 5GHz band.
- 802.11ad operates in the 60 GHz band.
- These standards provides data rate from 1Mb/s to 6.75Gb/s.

802.16-wiMAX:

- IEEE 802.16 is a collection of wireless broadband Standards, including extensive descriptions for the link layer (also called WiMax).

- **WiMax** standard provides a data rates from 1.5 Mb/s to 1Gb/s. The recent update (802.16m) provides data rates of 100Mb/s for mobile stations and 1Gb/s for fixed stations.

802.15.4 LR-WPAN:

- IEEE 802.15.4 is a collection of standards for **low rate wireless personal area network(LR-WPAN)**. These standards form the basis of specifications for high level communication protocols such as Zigbee. LR-WPAN standards provide data rates from 40 kb/s to 250kb/s. These standards provide low cost and low speed Communications for power constrained devices.

2G / 3G / 4G- Mobile communications:

- These are different generations of mobile communication standards including second generation (2G including GSM and CDMA), 3rd Generation (3G-including UMTS and CDMA2000) and 4th generation (4G- including LTE).
- IoT devices based on these standards can communicate over cellular networks.
- Data rates for these standards range from 9.6kb/s (for 2G) to upto 100Mb/s (for 4G).

Network / Internet layer :

- The network layers are responsible for sending of IP datagrams from the source network to the destination network.
- This layer Performs the host addressing and packet routing.

- The datagrams contains a source and destination address which are used to route them from the source to the destination across multiple networks.
- Host Identification is done using the hierarchical IP addressing schemes such as IPv4 or IPv6.
- **IPv4:** Internet protocol version 4(IPv4) is the most deployed internet protocol that is used to identify the devices on a Network using a hierarchical addressing scheme.
- IPv4 uses 32 bit addresses scheme that allows total of 2^{32} addresses. As more and more devices got connected to the internet these addresses got exhausted in the year 2011.
- The IPv4 has succeeded by IPv6.
- **IPv6:** It is the newest versions of internet protocol and successor to IPv4. IPv6 uses 128 bit address schemes that allows total of 2^{128} addresses.
- **6LoWPAN:** 6LoWPAN (IPv6 over low power wireless personal area networks) brings IP protocol to the low power device which have limited processing capability. It operate in the 2.4 GHz frequency range and provide the data transfer rate off to 250 kb/s.
- 6LoWPAN works with the 802.15.4 link layer protocol and defines compression mechanism for IPv6 datagrams over IEEE 802.15.4 based Network.

Transport layer :

- The Transport layer protocols provides end-to-end message transfer capability independent of the underlying network. The message transfer capability can be set up on connections, either using handshake (as in TCP) or without handshakes/ acknowledgements (as in UDP). It provides functions such as error control , segmentation, flow control and congestion control.
- **TCP:** Transmission Control Protocol(TCP) is the most widely used transport layer protocol, that is used by the web browsers (along with HTTP , HTTPS application layer protocols) email program (SMTP application layer protocol) and file transfer (FTP).
- TCP is a connection Oriented and stateful protocol. while IP protocol deals with sending packets, TCP ensures reliable transmissions of packets in order. TCP also provide error detection capability so that duplicate packets can be discarded and lost packets are retransmitted.
- The flow control capability of TCP ensures that the rate at which the sender sends the data is not too high for the receiver to process.
- **UDP:** unlike TCP, which requires carrying out an initial setup procedure, UDP is a connectionless protocol.
- UDP is useful for time sensitive application that have very small data units to exchange and do not want the overhead of connection setup.

- UDP is a transactions oriented and stateless protocol.
- UDP does not provide guaranteed delivery, ordering of messages and duplicate eliminations.

Application layer :

- Application layer protocol define how the application interfaces with the lower layer protocols to send the data over the network.
- The Application data are typically in files, is encoded by the application layer protocol and encapsulated in the transport layer protocol .Application layer protocol enable process-to-process connection using ports.
- Port numbers are used for application Addressing (e.g. Port 80 for HTTP, Port 22 for SSH etc)
- **HTTP:** Hypertext Transfer Protocol (HTTP) is the application layer protocol that forms the foundations of world wide web (WWW). HTTP includes commands such as GET, PUT, POST, DELETE, HEAD, TRACE, OPTIONS etc. The protocol follows a request-response model where the client sends request to server using the HTTP commands. HTTP is a stateless protocol and each HTTP request is independent of the other requests. An HTTP client can be a browser or an application running on the client (e.g. application running on an IoT device ,the mobile applications or other software.)
- HTTP protocol uses Universal Resource Identifiers (URIs) to identify HTTP resources.
- **CoAP:** Constrained application protocol is an application layer protocol for machine to machine (M2M) application.

- M2M meant for constrained environment with constrained devices and constrained networks. Like HTTP, CoAP is a web transfer protocol and uses a request- response model, however it runs on the top of the UDP instead of TCP. CoAP uses a client –server architecture where client communicate with server using connectionless datagrams. It is designed to easily interface with HTTP. Like HTTP, CoAP supports method such as GET, PUT, POST & DELETE .
- **WebSocket:** Websocket protocol allows full duplex communication over a single socket connections for sending message between client and server. WebSocket is based on TCP and Allows streams of messages to be sent back and forth between the client and server while keeping the TCP connection open. The client can be a browser, a mobile application and IoT device
- **MQTT** :Message Queue Telemetry Transport (MQTT) is a light-weight messaging protocol based on **public-subscribe model**. MQTT uses a client server Architecture where the clients (such as an IoT device) connect to the server (also called the MQTT broker) and publishes message to topics on the server.
- The broker forward the message to the clients subscribed to topic.
- MQTT is well suited for constrained environments where the devices have limited processing and memory resources and the Network bandwidth is low.
- **XMPP: Extensible Messaging and Presence Protocol(XMPP)** is a protocol for real-time communication and streaming XML data between network entities.XMPP powers wide range of applications including messaging, presence, data syndication, gaming multiparty chat and voice / voice calls. XMPP Allows sending small chunks of XML data from one network entity to another in near real time. XMPP supports both client to server and server to server communication paths.

- **DDS:** Data distribution service (DDS) is the data centric middleware standard for device-to-device or machine to machine communication. DDS uses a publish subscribe model where publishers (e.g. device that generate data) create topics to which subscribers(e.g. device that want to consume data) can subscribe. Publisher is an object responsible for data distribution and the subscriber is responsible for receiving published data. DDS provide quality of service (QoS) control and configurable reliability
- **AMQP:** Advanced Message Queuing protocols (AMQP) is an open application layer protocol for business messaging. AMQP support both point to point and publisher/subscriber models, routing and queuing.
- AMQP broker receive message from publishers (e.g. devices or applications that generate data) and route them over connections to consumers (application that process data) Publishers publish the message to exchanges which then distribute message copies to queues.

Logical design of IoT

- **Logical design of an IoT** system refers to an abstract representation of the entities and processes without going into low level specifics of the implementations .
- An IoT system comprises of a number of functional blocks that provide the system the capabilities for **identification, sensing, actuation ,communication and Management.**
- The functional blocks are---

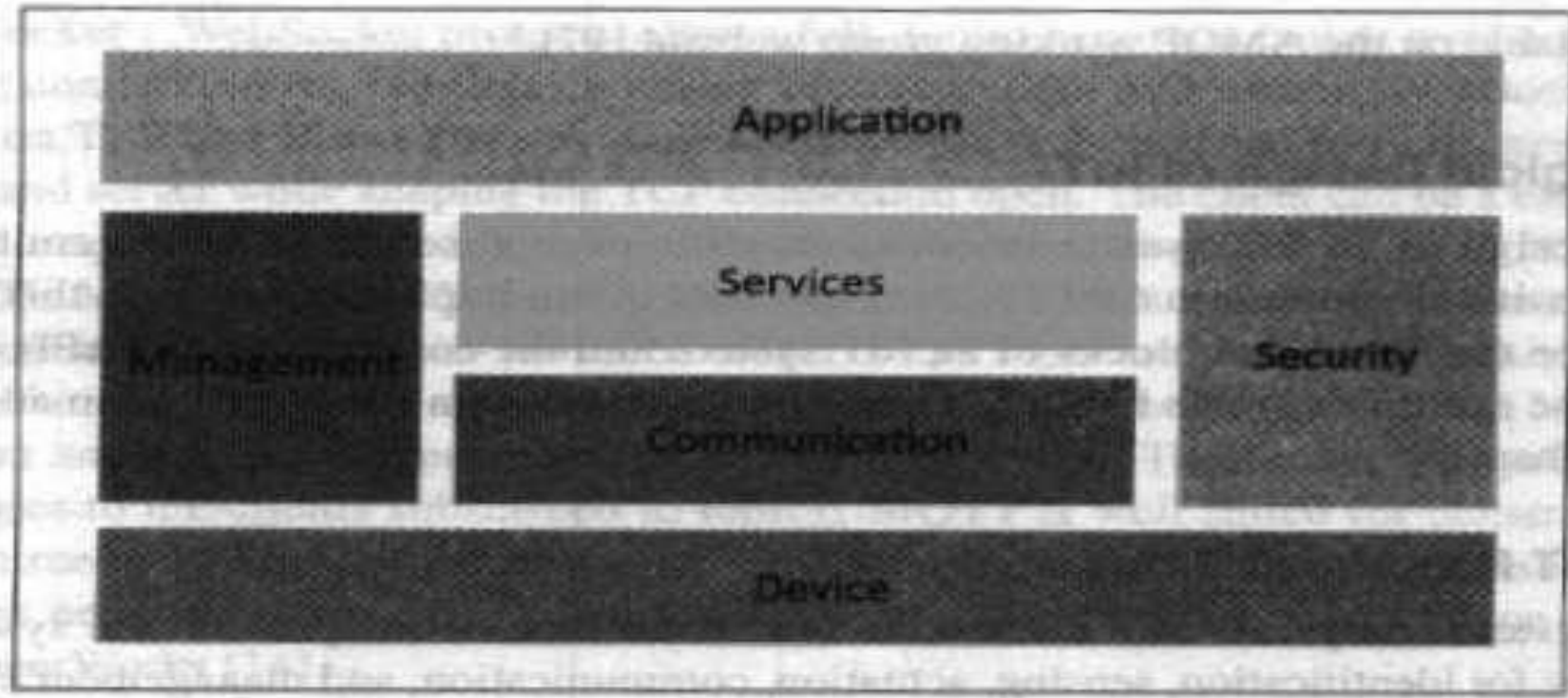


Figure 1.6: Functional Blocks of IoT

- **Devices:** An IoT system comprises of the devices that provide sensing, actuation, monitoring and control function.
- **Communication:** Communication block handle the communication for the IoT systems.

- **Services:** An IoT system uses various types of IoT services such as services for device monitoring, device control services, data publishing services and services for device Discovery.
- **Management:** Management Functional blocks provide various functions to govern the IoT system
- **Security:** Security functional block secures the IoT system and by providing functions such as authentication, authorization, message and content integrity and data security.
- **Application:** IoT application provides an interface that the users can use to control and monitor various aspects of the IoT system.
- Application also allow users to view the system status and view or analyse the processed data.

IoT communication models

- **Request response model:** Request-response is a Communications model in which the client sends request to the server and the server responds to the requests.
- when the server receives a request it decides how to respond, fetches the data, retrieves resource representations, prepares the response, and then sends the response to the client.
- Request-response model is a stateless communication model and each request response pair is independent of others. Figure 1.7 shows the client-server interactions in the request response model.

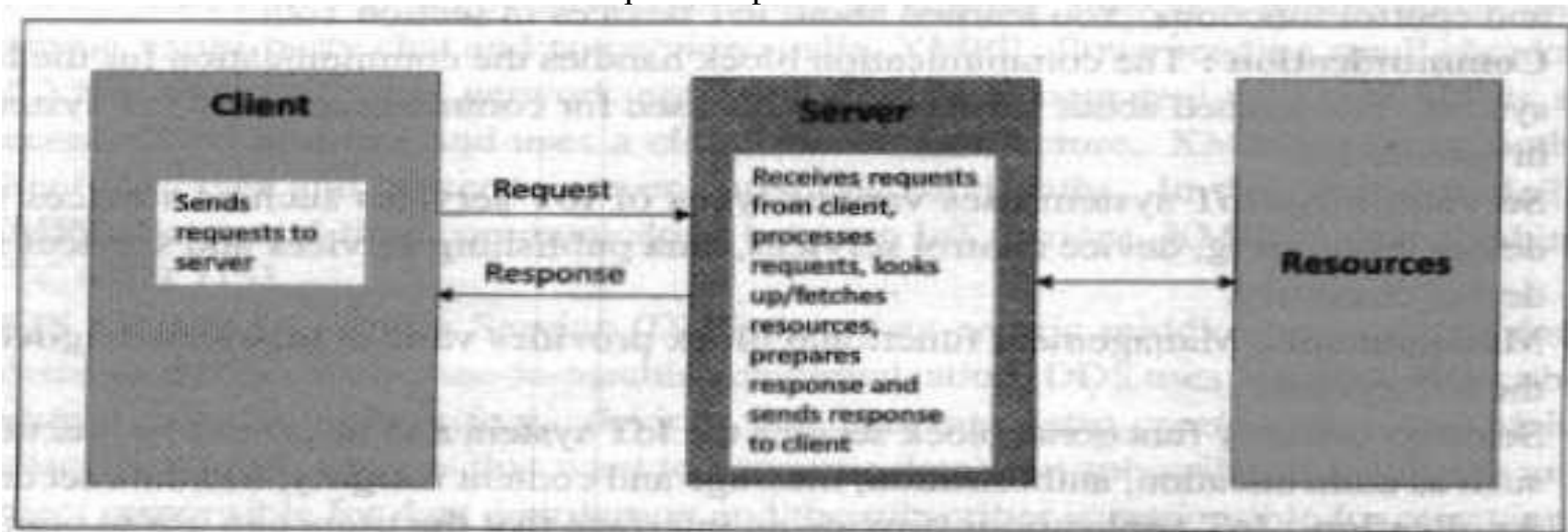


Figure 1.7: Request-Response communication model

- **Publish – Subscribe model:** Publish-Subscribe is a communication model that involve publishers, brokers and consumers.
- Publishers are the source of data. Publishers send the data to the topics, which are managed by the broker. Publishers are not aware of the consumer.
- **Consumers Subscribe** to the topics, which are managed by the broker. When the broker receives the data for a topic from the publisher, it sends the data to all the subscribed consumers. See fig 1.8 for this model.

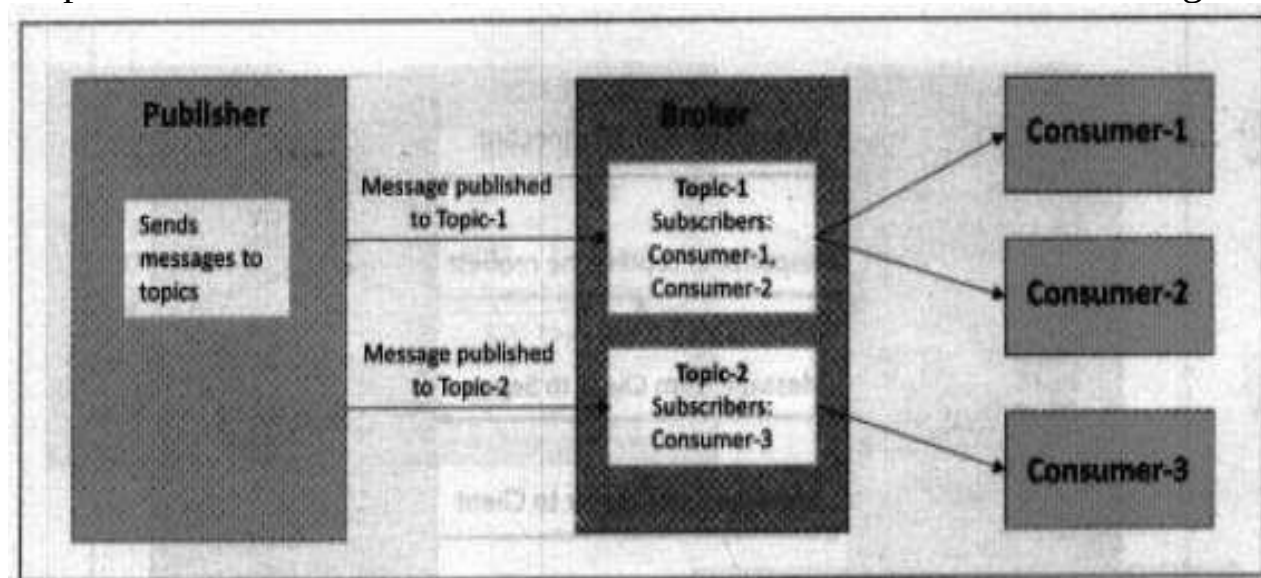


Figure 1.8: Publish-Subscribe communication model

- **Push pull model:** Push pull is communication model in which the data producers push the data to queues and the consumers pull the data from the queues.
- Producers do not need to be aware of the consumer.
- Queues help in decoupling the messaging between the Producers and Consumers .
- Queues also acts as a buffer which helps in situations when there is a mismatch between the rate at which the produces push data and the rate at which the consumers pull the data. See fig 1.9 for the model.

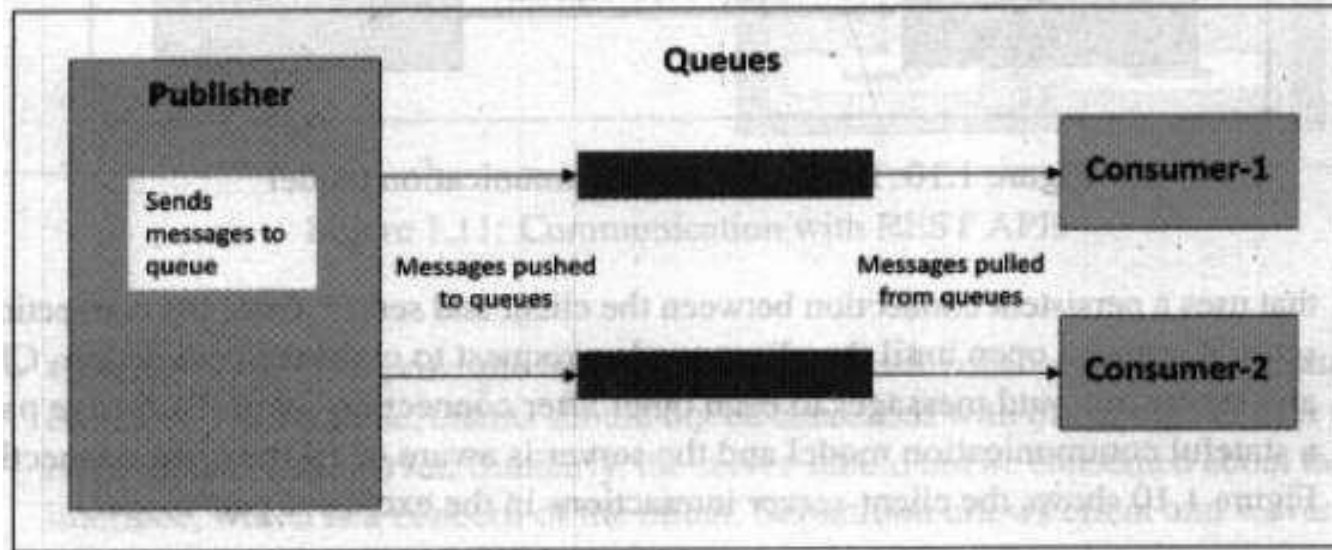
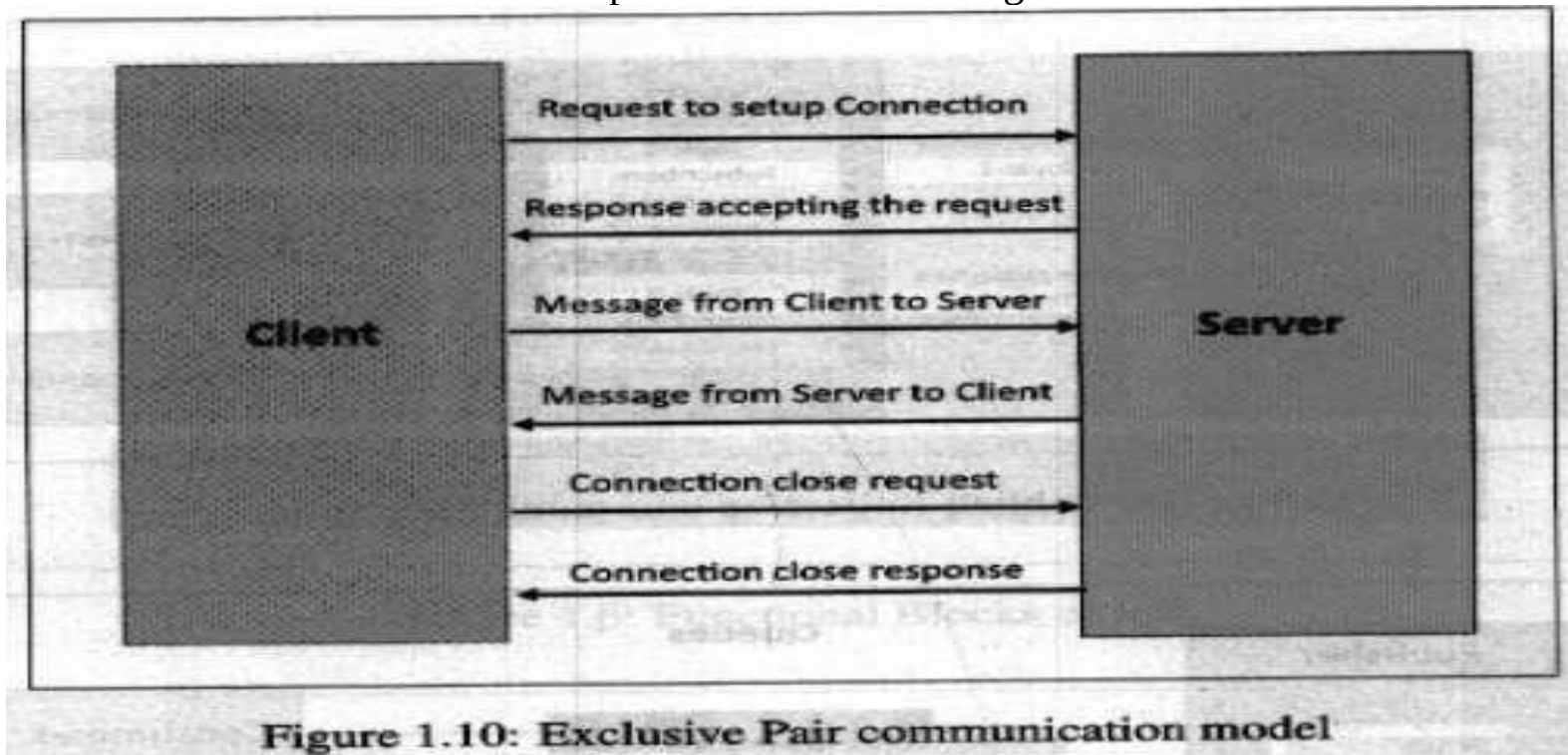


Figure 1.9: Push-Pull communication model

- **Exclusive pair:** Exclusive pair is a bi directional, fully duplex communication model that uses a persistent connections between the client and the server.
- Once the connection is setup it remains open until the client sends a request to close the connection.
- Client and server can send messages to each other after connection setup. Exclusive pair is a stateful Communication model and the server is aware of all the open connections. See fig 1.10 for the model



IoT communication Application Programming Interfaces (APIs)

- REST- based communication API:
- Representational state transfer (REST) is a set of architectural principles by which you can design web service and Web API that focus on a system's resources and how resources states are addressed and transferred. REST API follow the request-response communication model.
- The REST architectural constraints apply to the components, connectors, and data elements within a distributed hypermedia systems

REST architectural constraints

- Client server:
- The principle behind the client-server constraint is separation of concerns. For example, client should not be concerned with the storage of data which is a concern of the server.
- Similarly the server should not be concerned about the user interface which is a concern of the client.
- separation allows client and server to be independently developed and updated.

- **Stateless:** Each request from client to server must contain all the information necessary to understand the request , and cannot take advantage of any stored context on the server . The session state is kept entirely on the client.
- **Catch-able:** Catch constraint requires that the data within the response to a request be implicitly or explicitly labelled as catch-able or non-catch-able. If a response is catch-able then a client cache is given the right to reuse that response data for later, equivalent requests. Catching can partially or completely eliminate some attractions and improve efficiency and scalability.
- **Layered system:**
- Layered System constraint, constrains the behaviour of components such that each component cannot see beyond the immediate layer with which they are interacting.
- For example client cannot tell whether it is connected directly to the end server or to an intermediary along the way.
- System scalability can be improved by allowing intermediaries to respond to request instead of the end server, without the client having to do anything different.
- **Uniform interface:**
- Uniform interface constraints requires that the method of communication between client and server must be uniform.
- Resources are identified in the request (by URIs in the Web based System) and are themselves separate from the representation of the resources that are returned to the client.

- When client holds a representation of a resource it has all the information required to update or delete the resource (provided the client have required permissions).
- Each message includes enough information to describe how to process the message.
- **Code on demand:** Servers can provide executable code or scripts for clients to execute in their context.
- This is the only optional constraint

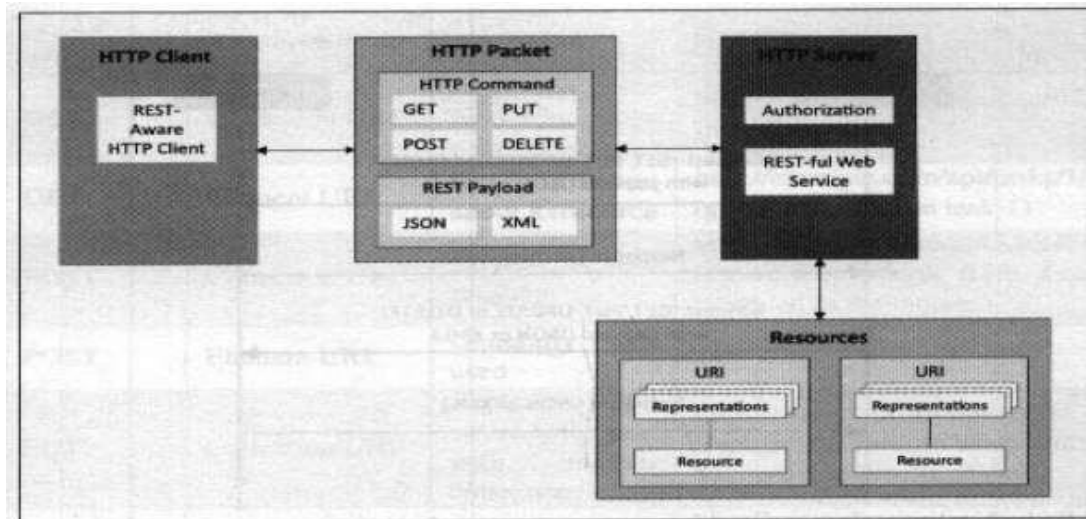


Figure 1.11: Communication with REST APIs

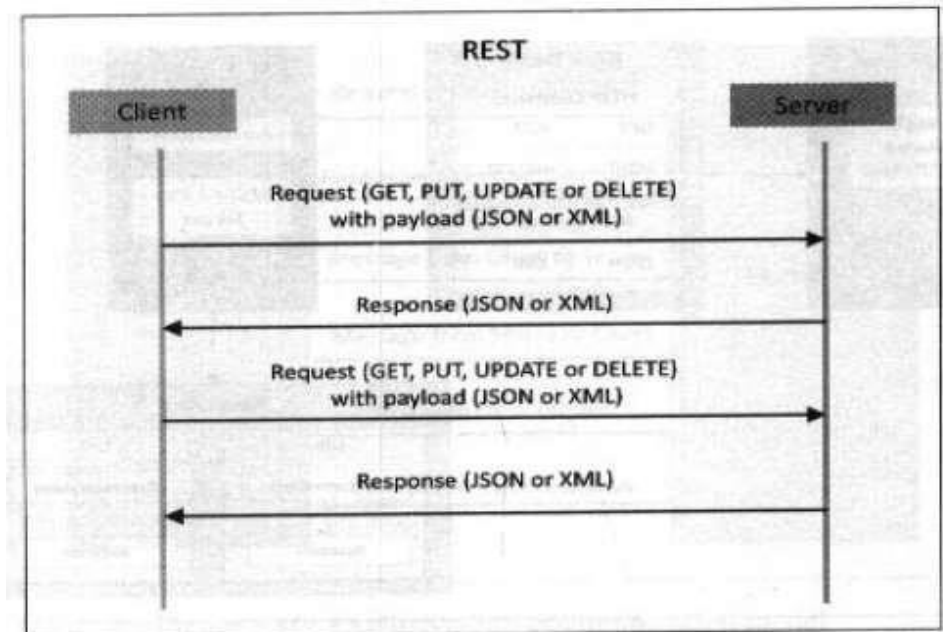


Figure 1.12: Request-response model used by REST

HTTP Method	Resource Type	Action	Example
GET	Collection URI	List all the resources in a collection	http://example.com/api/tasks/ (list all tasks)
GET	Element URI	Get information about a resource	http://example.com/api/tasks/1/ (get information on task-1)
POST	Collection URI	Create a new resource	http://example.com/api/tasks/ (create a new task from data provided in the request)
POST	Element URI	Generally not used	
PUT	Collection URI	Replace the entire collection with another collection	http://example.com/api/tasks/ (replace entire collection with data provided in the request)
PUT	Element URI	Update a resource	http://example.com/api/tasks/1/ (update task-1 with data provided in the request)
DELETE	Collection URI	Delete the entire collection	http://example.com/api/tasks/ (delete all tasks)
DELETE	Element URI	Delete a resource	http://example.com/api/tasks/1/ (delete task-1)

Table 1.1: HTTP request methods and actions

WebSocket based communication API:

- WebSocket API allow bi directional, full duplex communication between client and servers. WebSocket API follows exclusive pair communication model.
- Unlike request-response APIs such as REST, the WebSocket APIs allow full duplex communication and do not require new connection to be set up for each message to be sent.
- WebSocket communication begins with connection setup request send by the client to the server. This request (called as WebSocket handshake) is sent over HTTP and the server interprets it as an upgrade request.
- If the server supports WebSocket Protocol, the server responds to the WebSocket handshake response. After the connection is setup the client and the server can send data/messages to each other in full duplex mode.
- WebSocket API reduce network traffic and latency as there is no overhead for connection setup and termination requests for each message.
- WebSocket is suitable for IoT applications that have low latency or high throughput requirements

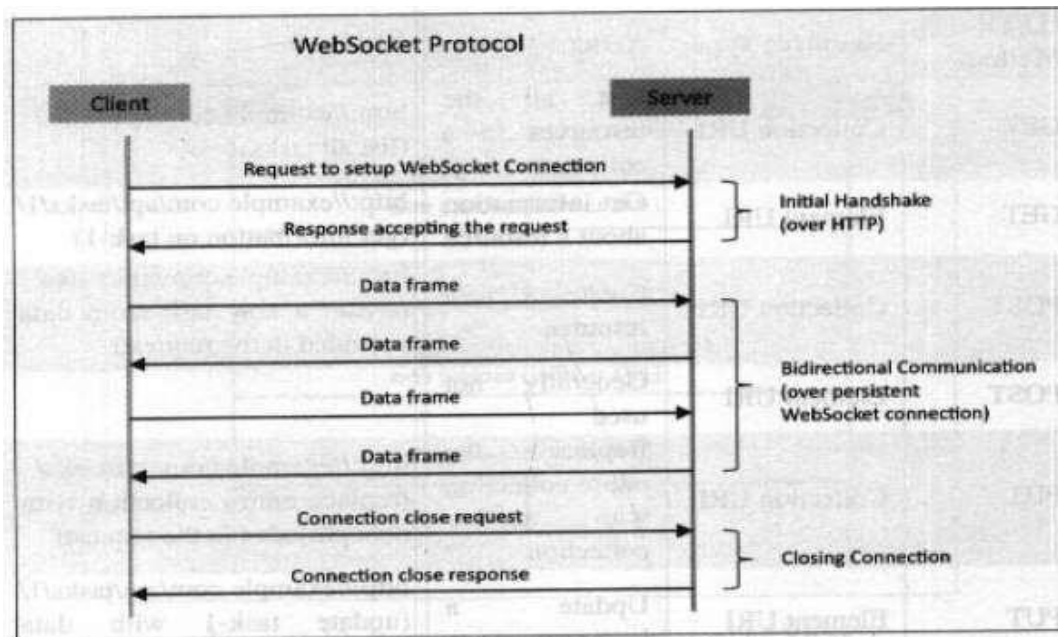


Figure 1.13: Exclusive pair model used by WebSocket APIs